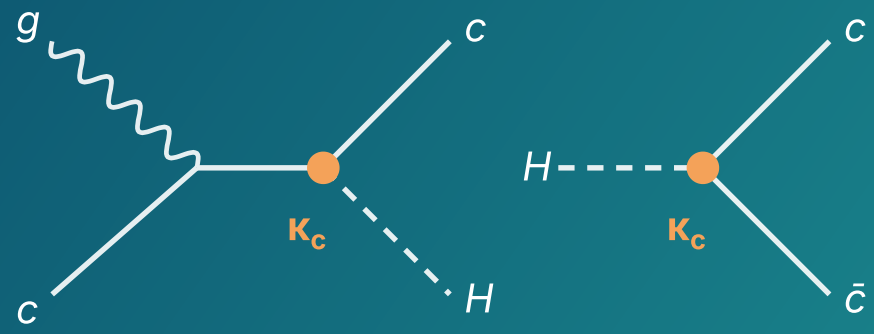
Constraints on Higgs–charm couplings at CMS

Higgs 2026 · Louvain-la-Neuve · 28 October 2026

Chirayu Gupta
Vrije Universiteit Brussel

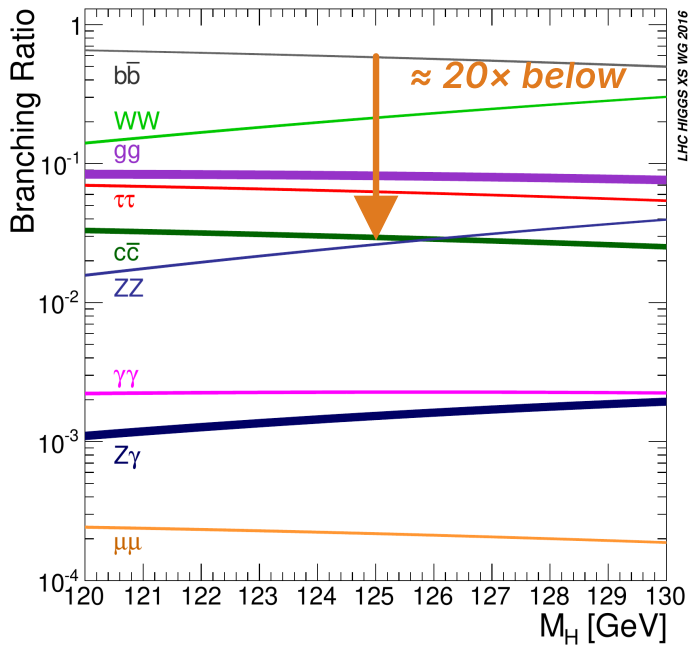
on behalf of the CMS Collaboration



Why the Higgs–charm coupling?

Second-generation quark Yukawas are the open frontier

- Since 2012: couplings to **t, b, τ** measured, all SM-like
- Evidence for **H → μμ** (backup)
- **Charm is the most accessible second-generation quark**



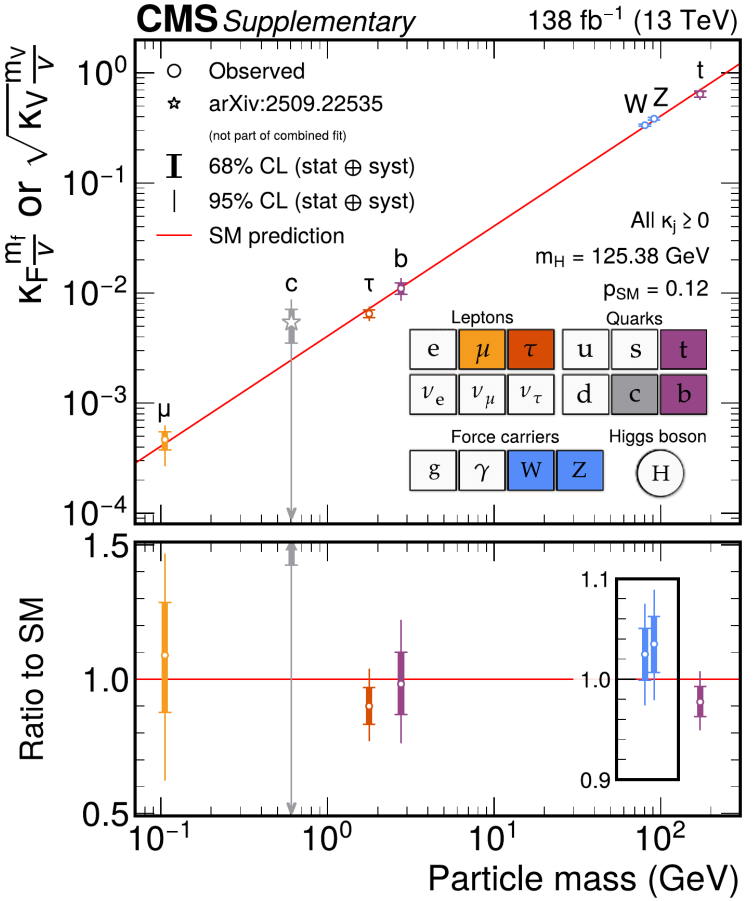
$\mathcal{B}(H \to c\bar{c}) \approx$ **2.9%**, about **20× below** $b\bar{b}$, with huge QCD backgrounds and hard c tagging (slide 3)

$$\kappa_c = y_c / y_c^{SM} \quad \text{(SM: 1)}$$

Flavour-compatible BSM, e.g. spontaneous flavour violation, allows $|\kappa_c| \sim 3$: **current sensitivity is entering this region**

SFV: PRL 123 (2019) 031802
κ_c in SFV: JHEP 02 (2025) 067

LHC Higgs XS WG, Yellow Report 4, arXiv:1610.07922



CMS, Rep. Prog. Phys. 89 (2026) 087801 (suppl.); κ_c point: separate $t\bar{t}H$+VH result, not in the fit

➜ *Does the Higgs give charm its mass?*

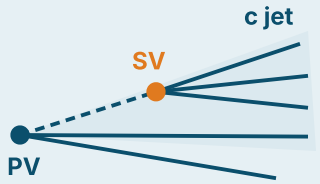
CMS-DP-2025-081

Charm tagging

Finding c jets between b and light jets

Charm signature

c-hadron mass and lifetime lie **between light and b**: a secondary vertex (SV) like b jets, but closer to the primary vertex, so **easily mistaken for light jets**



Algorithm evolution at CMS

Likelihood / BDT

JP
track impact-parameter likelihood

CSV, CSVv2
likelihood ratio → MLP

Run 2 c tagger
BDT: CvsL, CvsB

Deep NN

DeepCSV
dense DNN (4 × 100)

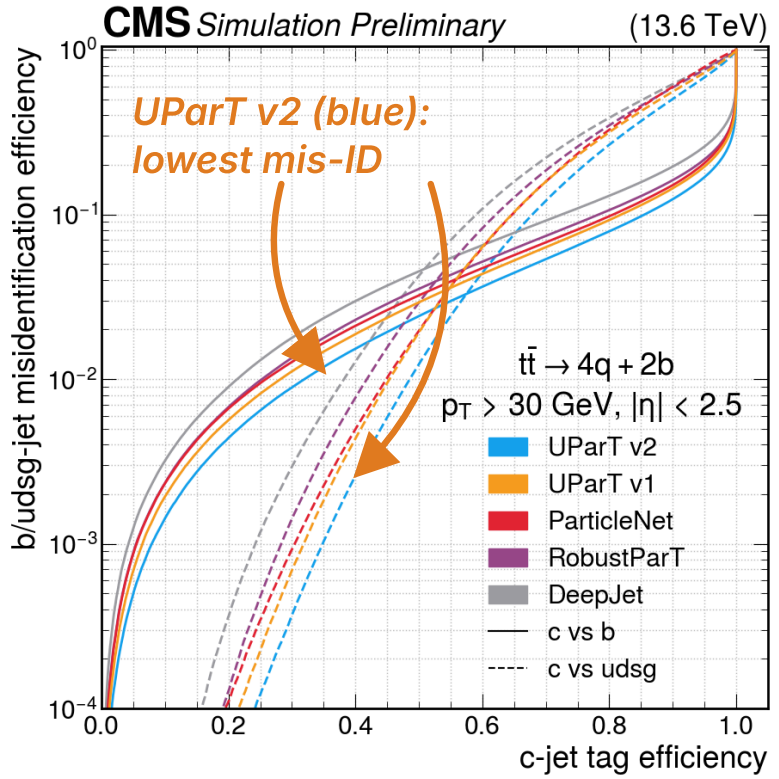
DeepJet
conv + LSTM on constituents, SVs

Particle cloud

ParticleNet
dynamic graph CNN

UParT
transformer + pairwise features

Run 1 → Run 3: likelihoods and trees → dense nets → graphs and attention



$t\bar{t}$ simulation, 13.6 TeV · solid: c vs b, dashed: c vs light

In this talk **DeepJet · VH resolved, cH** **ParticleNet · VH merged, $t\bar{t}$H** **DeepDoubleX · boosted H**

How CMS probes κ_c

In the decay, in production, and through the Higgs width

Direct H → $c\bar{c}$

slides 5–11

rate ∝ κ_c^2/Γ_H · VH, $t\bar{t}H$, boosted (mostly ggH)

$|\kappa_c| < 3.5$ (2.7) $t\bar{t}H + VH$

PRL 131 (2023) 061801 · PRL 136 (2026) 011801 · PRL 131 (2023) 041801

H + charm production

slides 12–14

cg → cH: κ_c in the production vertex · H → γγ, WW

$|\kappa_c| < 47$ (51)

JHEP 11 (2025) 060 · JHEP 08 (2026) 051

Indirect: Higgs rates and width

slide 15

on-shell and **new** off-shell H → ZZ → 4ℓ; range depends on assumptions

$\kappa_c \in [-3.3, 2.8] \ldots [-8.6, 7.5]$

PRD 112 (2025) 112001 · arXiv:2607.23352

Rare exclusive decays

slide 16

H → J/ψ γ, ψ(2S) γ: sensitive to the *sign* of κ_c

$-166 < \kappa_c < 208$

PLB 865 (2025) 139462

$\mu = \sigma\mathcal{B}/(\sigma\mathcal{B})_{SM}$ · Γ_H = total Higgs width · μ limits and κ_c intervals at 95% CL, observed (expected)

→ All κ_c results: Run 2 (13 TeV, up to 138 fb^{-1}) · Run 3 so far: better triggers (slide 17)

VH(H → cc̄): strategy

PRL 131 (2023) 061801 · CMS-HIG-21-008

Cleanest signature: the W or Z decays to leptons

0ℓ · Z → νν | **1ℓ · W → ℓν** | **2ℓ · Z → ℓℓ**

Main backgrounds V+jets and tt̄, normalised in control regions

below **$p_T(H_{cand})$ = 300 GeV** *above*

Resolved · two small-R jets (R = 0.4)

- **>95% of the signal**, but more background
- **DeepJet** CvsL, CvsB: the two highest-CvsL jets form H_{cand}
- Mass peak up to **20% sharper**: c-jet energy regression, FSR-jet recovery, kinematic fit (2ℓ)

Fit: BDT score (m(H_{cand}) among the inputs)

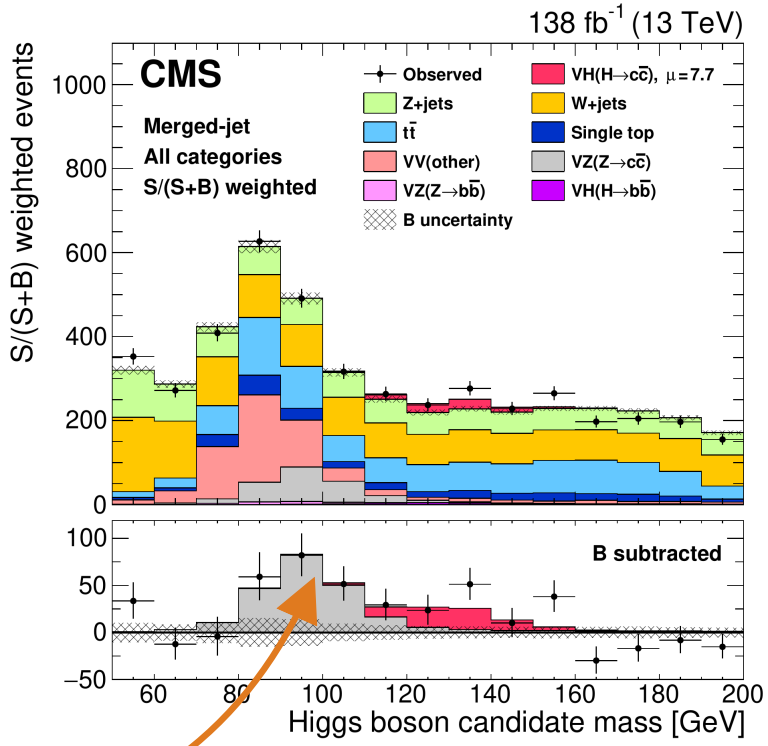
Merged · one large-R jet (R = 1.5)

- **<5% of the signal**, but much less background
- **ParticleNet** cc̄ tagger (graph NN, mass-decorrelated): **4–7×** better rejection than DeepAK15
- ParticleNet mass regression: **≈50% better** mass resolution

Fit: m(H_{cand}) (BDTs decorrelated from it)

Validation: the same analysis sees **VZ(Z → cc̄) at 5.7σ** (grey peak, right): c tagging and background model work → backup

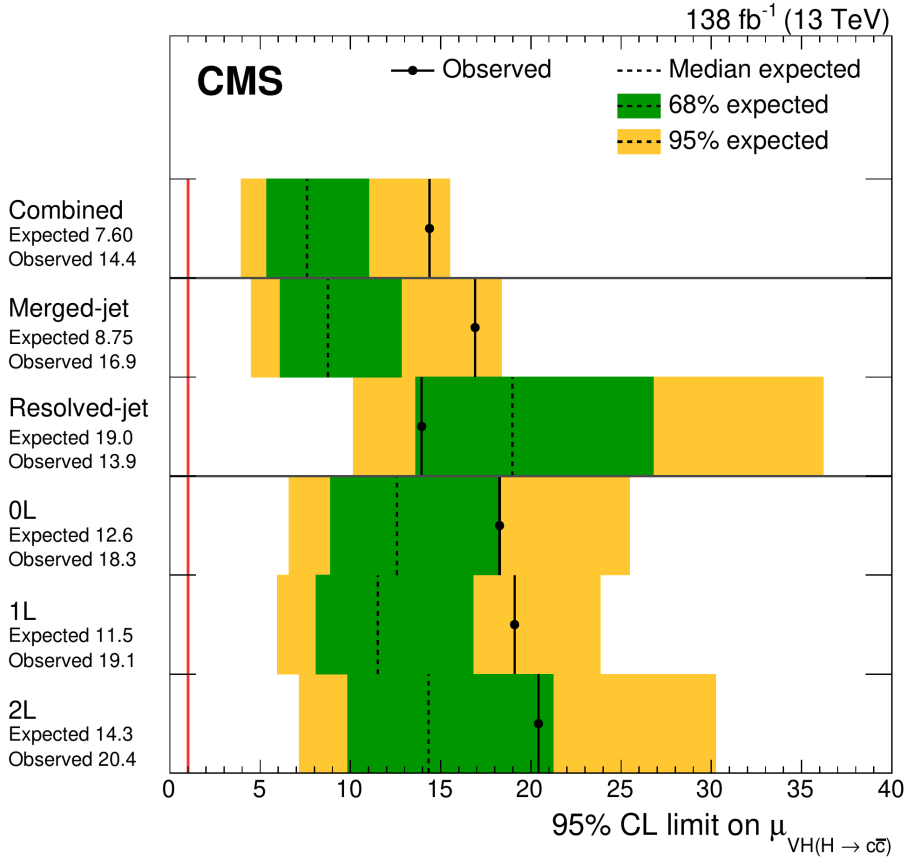
Merged-jet mass; bottom: VZ (grey, ≈ 90 GeV) and VH (red) after background subtraction



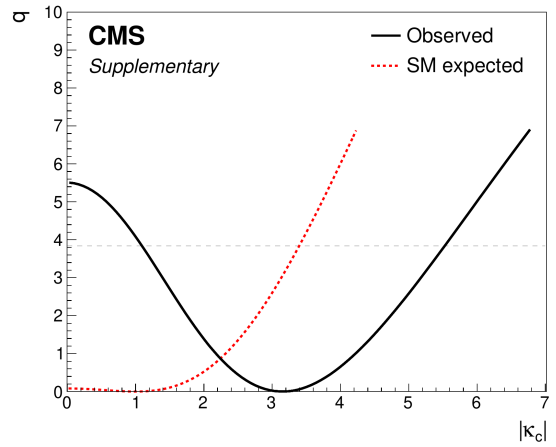
VH(H → $c\bar{c}$): results

PRL 131 (2023) 061801 · CMS-HIG-21-008

Best expected μ of any single Run 2 channel



95% CL limits on μ per category · red line: SM (μ = 1) · the **merged-jet** category carries most of the sensitivity



q = −2Δln L vs $|\kappa_c|$ (supplementary) · 95% CL where q < 3.84 (dashed line)

95% CL · observed (expected)

μ < 14 (7.6) **1.1 < $|\kappa_c|$ < 5.5** (< 3.4)

μ limits with CLs; κ_c interval from the profile-likelihood scan

85%

of the uncertainty is statistical

Statistics-limited: more data helps directly

Statistical	**85%**
Experimental	**48%**
Theory	**22%**

share of the total uncertainty on μ, in quadrature · details → backup

➜ Expected μ < 7.6: the best of any single Run 2 channel

PRL 131 (2023) 041801 · CMS-HIG-21-012

Boosted H → $c\bar{c}$

Gluon-fusion enriched: one large-radius jet at high p_T

① Boosted regime: $p_T^H > 450$ GeV

- High p_T suppresses the QCD background
- H → $c\bar{c}$ in **one large-radius jet** (R = 0.8); soft drop grooms it and sharpens the mass m_{SD}
- Signal: 55% ggF, 25% VBF, 20% VH

② DeepDoubleX tagger

- Convolutional + recurrent network on particle-flow candidates and secondary vertices
- **Passing** region (signal-rich) and **failing** region (background-rich)
- Mass-decorrelated: QCD in the passing region is **predicted from data** in the failing region

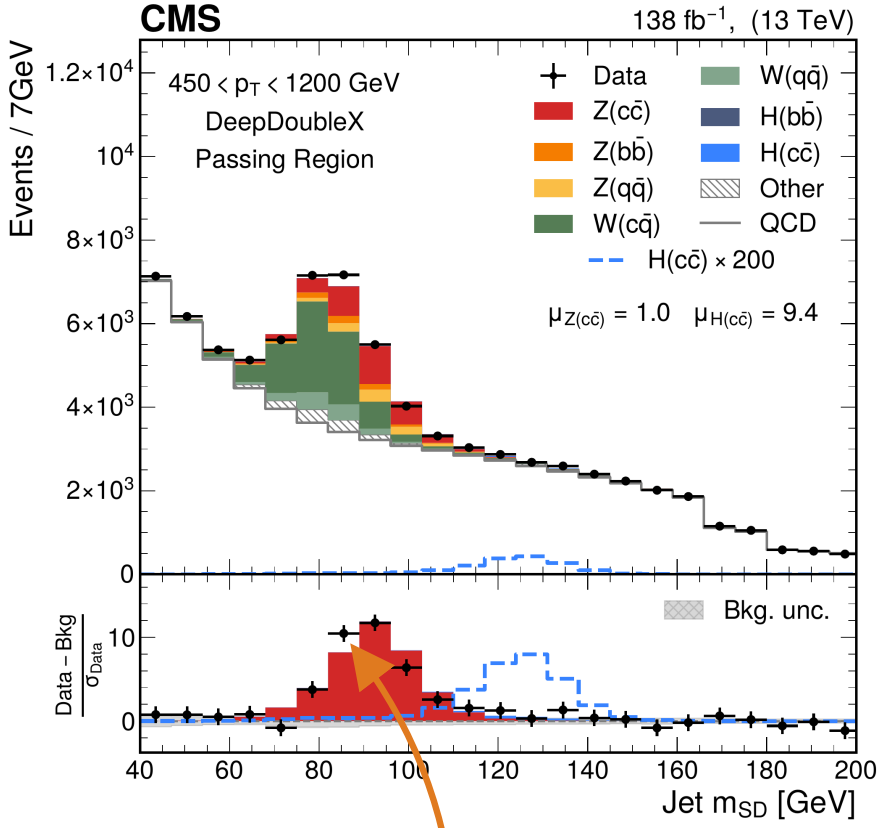
③ Statistical analysis

- Simultaneous binned fit of m_{SD} in both regions, in 6 p_T bins
- Upper limit from the profile likelihood ratio (CLs)

$$\sigma\mathcal{B} / (\sigma\mathcal{B})_{SM} < 47\ (39)$$

95% CL · observed (expected) · no κ_c interpretation in the paper

Jet mass in the passing region · H → $c\bar{c}$ dashed (×200)



Z → $c\bar{c}$ observed for the first time in Z+jets, well over 5σ

$t\bar{t}H(H \to c\bar{c})$: selection and jet tagging

PRL 136 (2026) 011801 · CMS-HIG-24-018

Three $t\bar{t}$ decay channels, every jet flavour-tagged

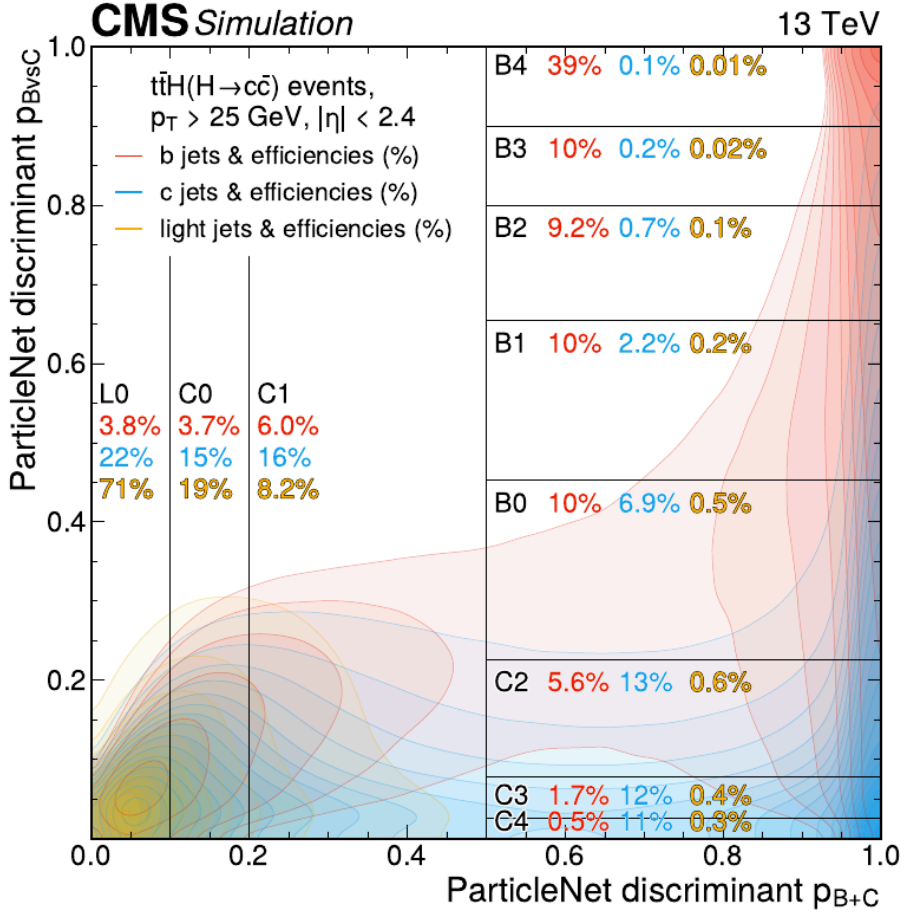
0ℓ — **≥ 7 jets**, H_T > 500 GeV
fully hadronic $t\bar{t}$

1ℓ — **1 lepton + ≥ 5 jets**
single-lepton $t\bar{t}$

2ℓ — **2 opposite-charge leptons + ≥ 4 jets**, Z veto
dilepton $t\bar{t}$

ParticleNet (graph NN) tags every jet as b, c or light: **11 categories** (right), up to **2×** better background rejection than DeepJet

Main background: $t\bar{t}$+jets, split by the flavour of the extra jets: $t\bar{t}$+b, +≥2b, +c, +≥2c, +light



b, c and light jets in the ParticleNet (b+c vs light, b vs c) plane, with the 11 tagging categories

$t\bar{t}H(H \to c\bar{c})$: event classifier and regions

PRL 136 (2026) 011801 · CMS-HIG-24-018

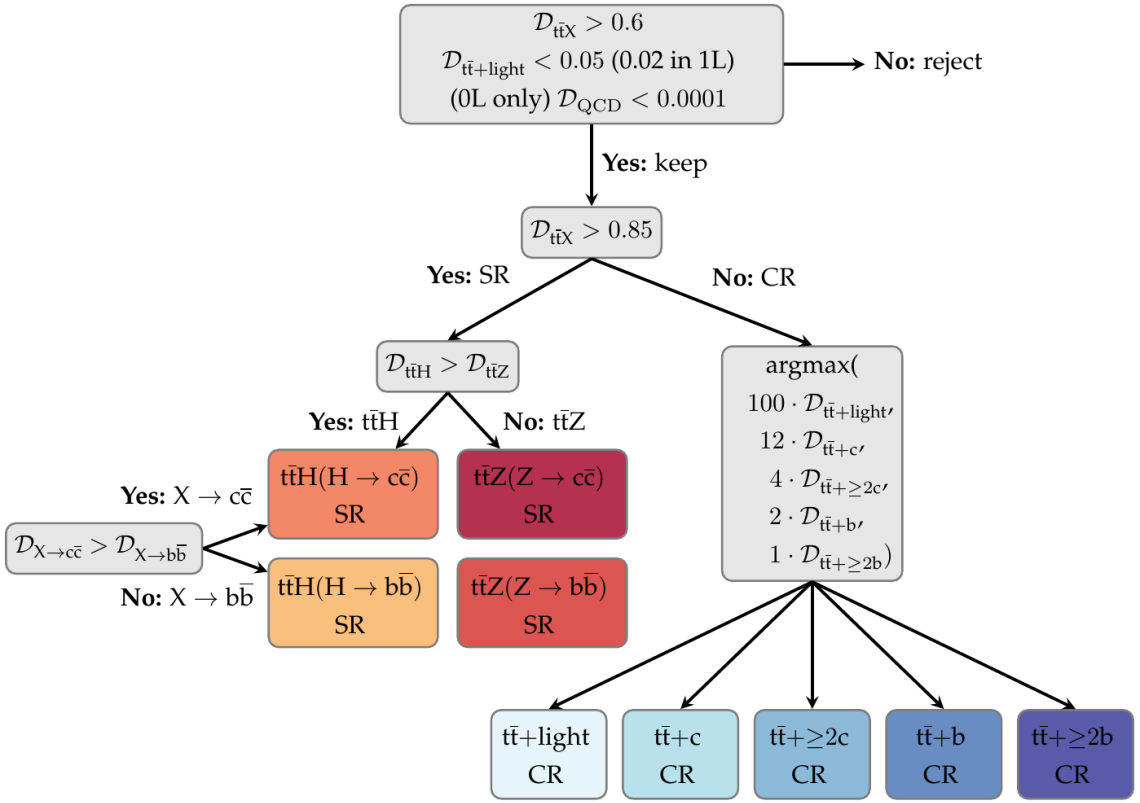
Signal, control and validation regions from one classifier

Event classifier: ParT (transformer) on jets, leptons and p_T^{miss}: one score per process (4 signals, 5 $t\bar{t}$+jets, + QCD in 0ℓ)

$D_{t\bar{t}X}$ = sum of the four signal scores

rejected	Validation	5 control regions	4 SRs
< 0.4	0.4–0.6	0.6–0.85	> 0.85

- **SRs**: $t\bar{t}H$ / $t\bar{t}Z$ × $c\bar{c}$ / $b\bar{b}$; c-enriched SRs need ≥ 2 c jets
- **CRs**: one per $t\bar{t}$+jets flavour; their normalisations float in the fit
- **Validation**: same structure, signal-depleted: good agreement → backup
- **QCD** (0ℓ): a classifier cut leaves < 1% of $t\bar{t}$+jets
- **Fit**: ParT score in all regions of the 3 channels at once

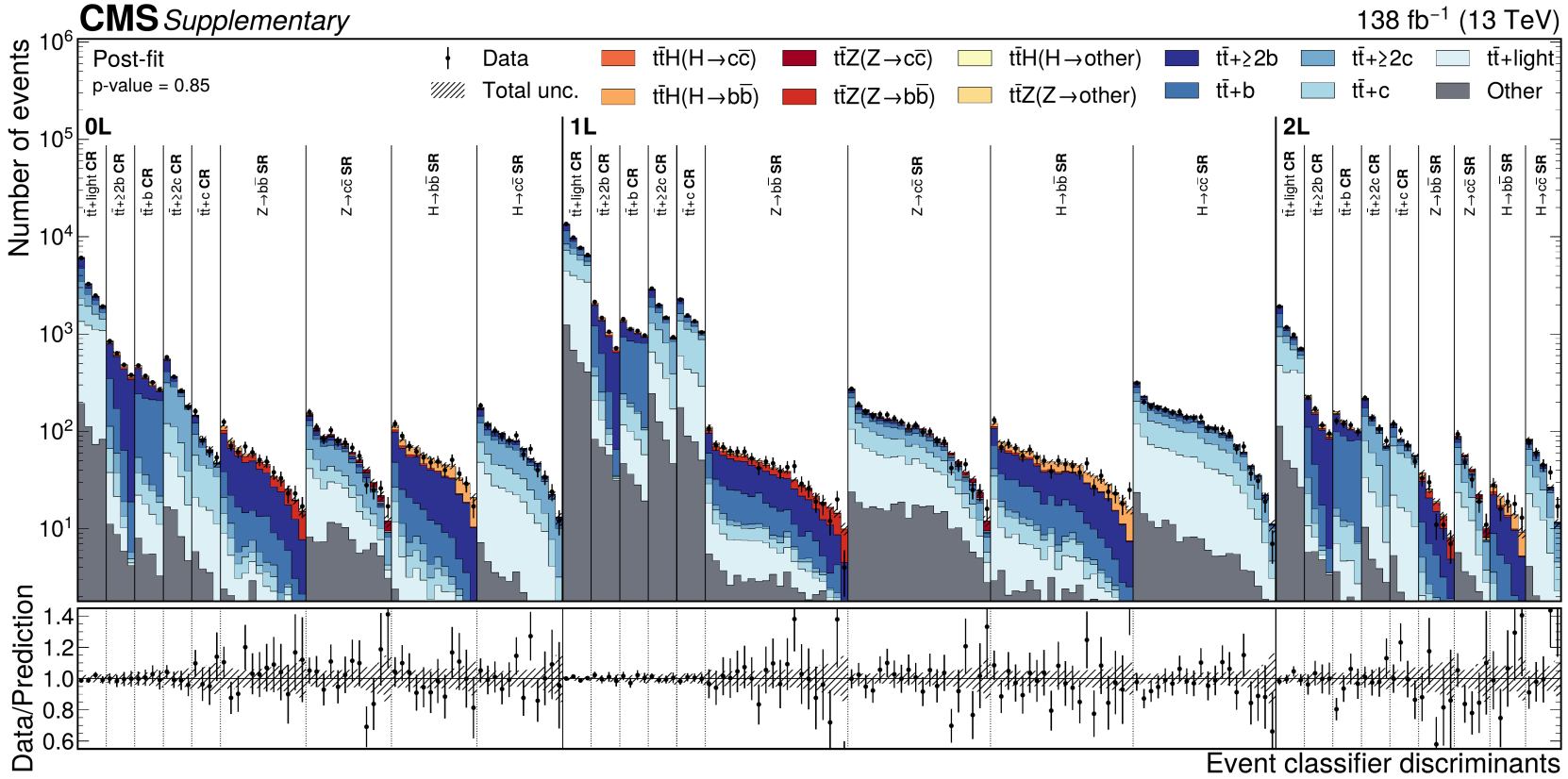


Categorisation of events with $D_{t\bar{t}X} > 0.6$ (same tree in the VR, 0.4–0.6)

$t\bar{t}H(H \to c\bar{c})$: results

PRL 136 (2026) 011801 · CMS-HIG-24-018

One fit to all regions: four signals and every $t\bar{t}$+jets component



Post-fit ParT discriminants in the 5 CRs and 4 SRs of each channel (0ℓ, 1ℓ, 2ℓ) · fit p-value 0.85

$t\bar{t}H(H \to c\bar{c})$ · 95% CL · observed (expected)

$\mu < 7.8$ (8.7)

$|\kappa_c| < 3.0$ (3.3)

$\kappa_b = 1$ · with κ_b fitted: $|\kappa_c| < 4.2$ obs. → backup

	best-fit μ	obs. (exp.)
$t\bar{t}H(b\bar{b})$	$0.91\ ^{+0.26}_{-0.22}$	4.4σ (4.5σ)
$t\bar{t}H(c\bar{c})$	**−1.6 ± 4.5**	–
$t\bar{t}Z(b\bar{b})$	$1.47\ ^{+0.45}_{-0.41}$	3.5σ (2.4σ)
$t\bar{t}Z(c\bar{c})$	$1.02\ ^{+0.79}_{-0.84}$	1.2σ (1.3σ)

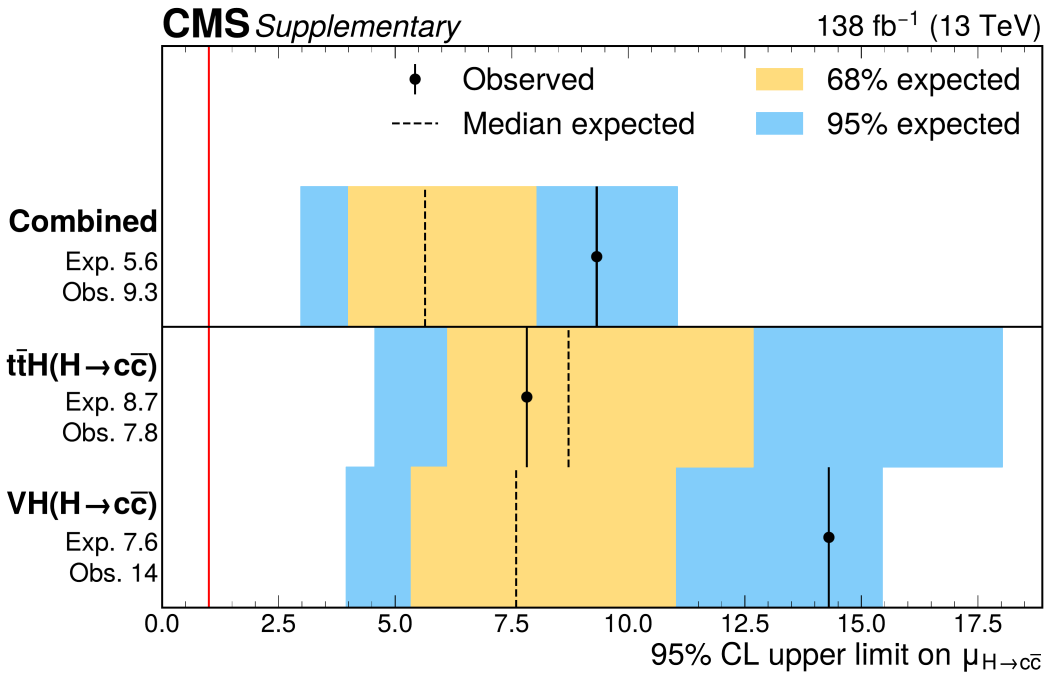
$t\bar{t}Z(c\bar{c})$ and $t\bar{t}Z(b\bar{b})$ are the standard candles: both consistent with the SM · limits per channel and event yields → backup

→ A second direct channel, with expected sensitivity comparable to VH

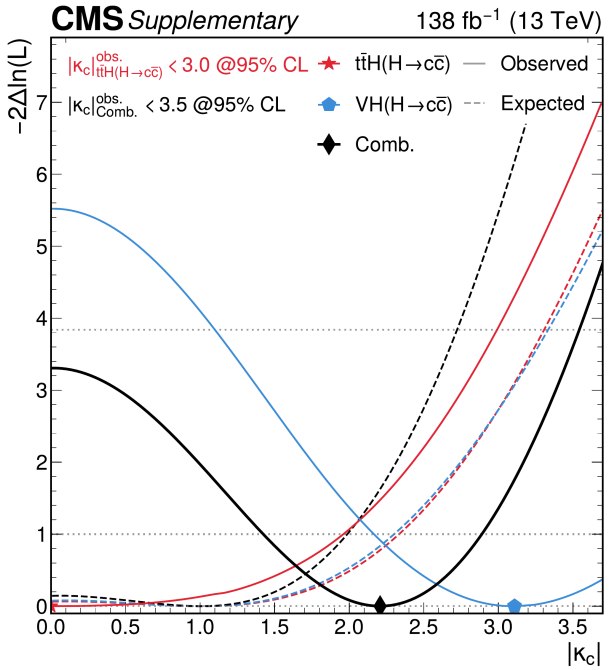
$t\bar{t}H$ + VH combination

PRL 136 (2026) 011801 · CMS-HIG-24-018

Best expected direct sensitivity to κ_c



95% CL limits on $\mu_{H\to c\bar{c}}$: $t\bar{t}H$, VH and combined



$-2\Delta\ln L$ vs $|\kappa_c|$ ($\kappa_b = 1$) · red $t\bar{t}H$, blue VH, black comb.

$t\bar{t}H$ + VH · 95% CL
observed (expected)

$\mu < 9.3$ (5.6)

$|\kappa_c| < 3.5$ (2.7)

SM production, $\kappa_b = 1$

26%
better expected μ limit than VH alone

36%
better expected μ limit than $t\bar{t}H$ alone

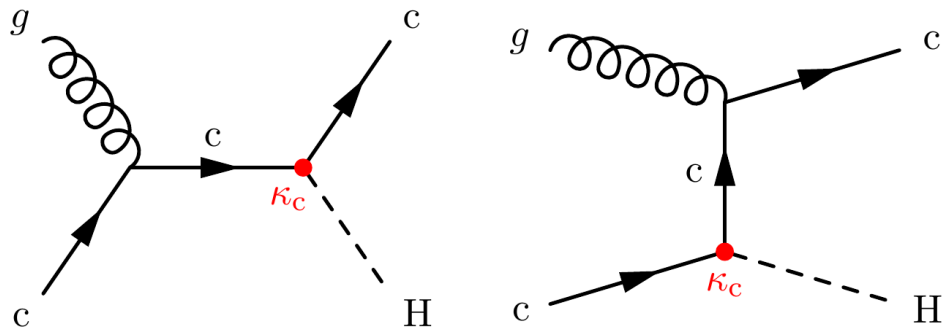
$|\kappa_c| < 2.7$
expected, below the BSM-motivated ~3

➜ The two direct channels together set the best expected constraint on κ_c

H + charm production

JHEP 11 (2025) 060 · JHEP 08 (2026) 051

κ_c in the production vertex, independent of the H → $c\bar{c}$ decay



LO diagrams for cH production; κ_c at the marked vertices

Signal: a clean Higgs decay (γγ or WW) + one c-tagged jet

Challenges

1 Tiny rate

σ_{cH} = 90 fb, about 0.2% of all Higgs production: $\sigma\mathcal{B} \approx$ **0.2 fb** (γγ), ≈ 20 fb (WW)

2 Yukawa-insensitive processes

ggH and other Higgs processes with an extra c jet (e.g. from g → $c\bar{c}$) give the **same final state** without involving κ_c

3 b-associated Higgs (bH)

σ_{bH} = 660 fb, 7× larger: b jets mistagged as c, so c-vs-b separation matters

4 Modelling

Flavour scheme for cH (4FS vs 3FS): **25–30%** uncertainty; heavy-flavour modelling of ggH (T. Bevilacqua, next talk)

→ A complementary handle on κ_c, but small rates and irreducible H+c make it hard

JHEP 11 (2025) 060 · CMS-HIG-23-010

cH(H → γγ)

First search for H + charm in the diphoton channel, 138 fb^{-1}

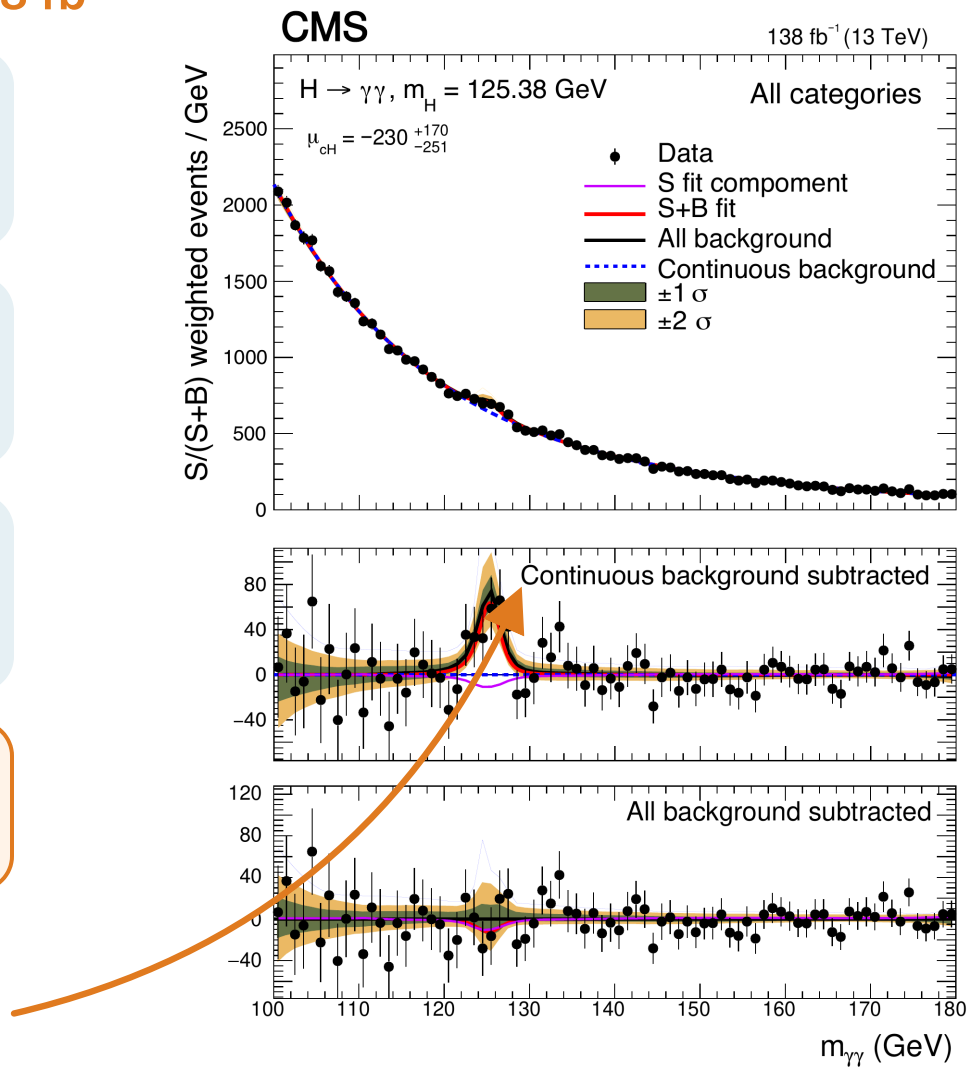
① Selection
- Two identified photons
- Leading jet (p_T > 25 GeV) **c-tagged** with DeepJet

② Two BDTs, no c-tag inputs
- cH vs **ggH**, the largest resonant background
- cH vs the **continuum** γγ + jets

③ Categories and fit
- 27 categories (9 per year)
- Signal from fits to $m_{\gamma\gamma}$

μ_{cH} < 243 (355) **$|\kappa_c|$ < 38** (73)

95% CL · observed (expected)



The 125 GeV peak is explained by the other Higgs processes (mostly ggH)

→ $|\kappa_c| \lesssim 40$: $\sigma\mathcal{B} \approx 0.2$ fb, and Yukawa-insensitive H+c looks the same

cH(H → WW) and the γγ + WW combination

JHEP 08 (2026) 051 · CMS-HIG-24-009

evμν + a c jet, then combined with γγ

① Selection

- One electron + one muon, p_T^{miss} > 20 GeV
- ≥ 1 **c jet** (DeepJet): 43% c efficiency, 27% b and 5% light mistag, to separate cH from bH

② Two BDTs

- vs non-Higgs backgrounds
- vs other Higgs processes; signal regions with 1 or > 1 c jets

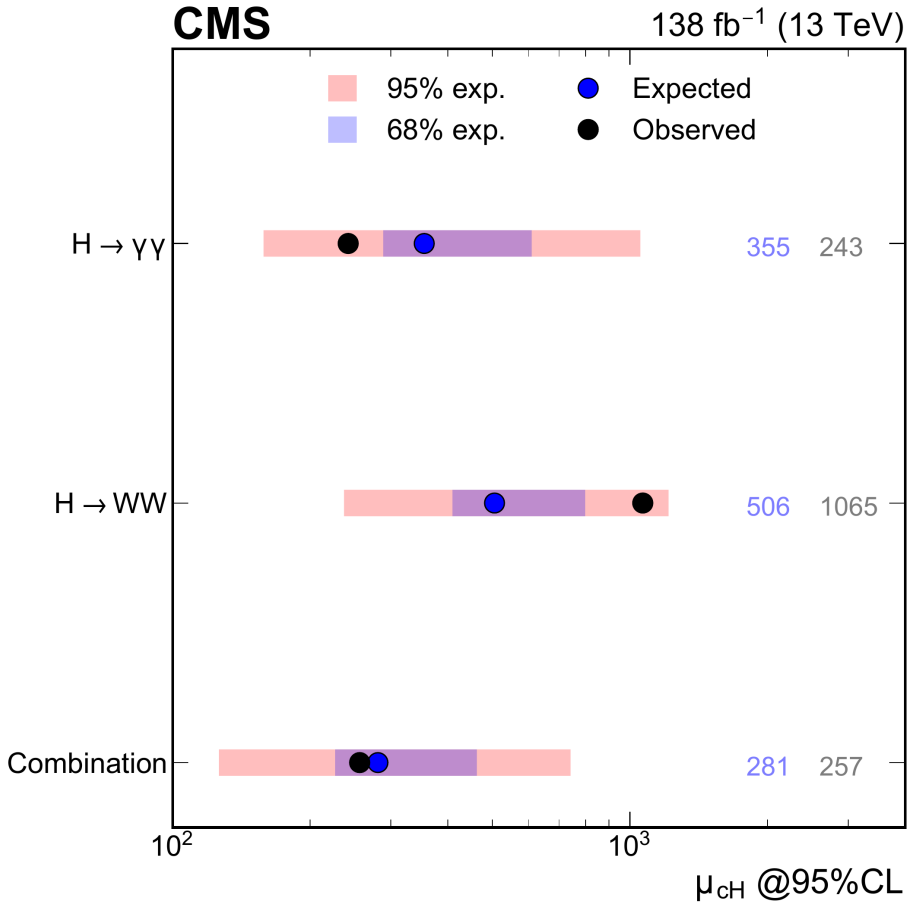
③ Fit

- μ_{cH} is the only parameter; all other Higgs processes fixed to their SM prediction (floating the H+c part → backup)

γγ + WW · 95% CL · observed (expected)

μ_{cH} < 257 (281) $|\kappa_c|$ < 47 (51)

WW alone: μ_{cH} < 1065 (506), $|\kappa_c|$ < 211 (95)



μ_{cH} limits: γγ, WW and combined

→ κ_c from production alone, independent of the H → $c\bar{c}$ decay; statistics-limited

Indirect constraints on κ_c

PRD 112 (2025) 112001 · arXiv:2607.23352

The Higgs width from on- and off-shell H → ZZ → 4ℓ

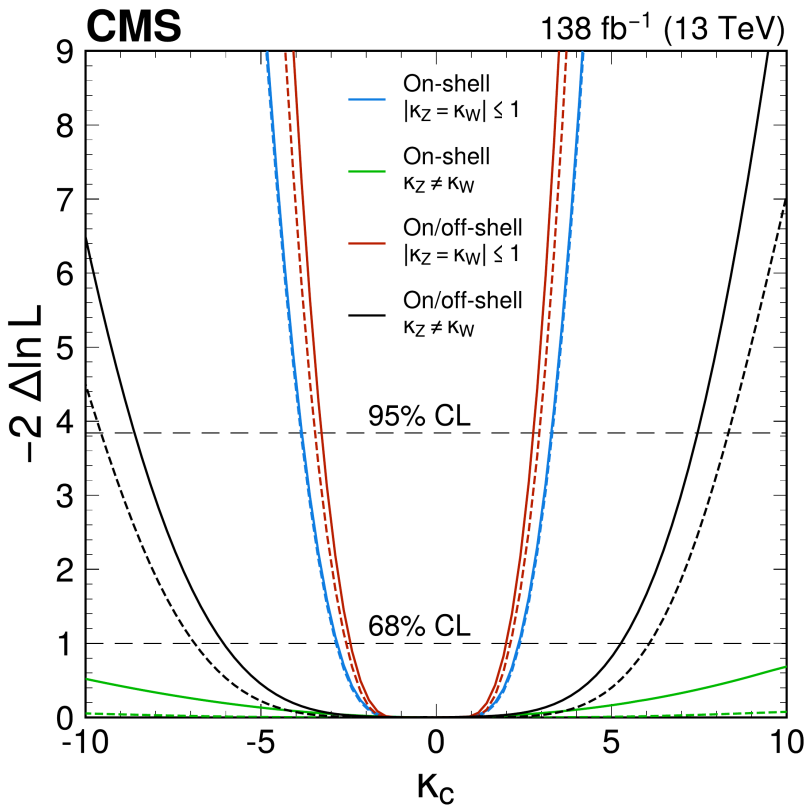
$$\sigma_{on} \propto \frac{g_p^2\, g_d^2}{\Gamma_H} \qquad \sigma_{off} \propto g_p^2\, g_d^2$$

g_p, g_d: production and decay couplings ($g_d \propto \kappa_Z$)

1. **Large κ_c** adds $c\bar{c}$ → H production and a larger width Γ_H
2. **On-shell** $\propto 1/\Gamma_H$: larger κ_Z can hide larger Γ_H ⇒ need $|\kappa_Z| \leq 1$
3. **Off-shell** has no Γ_H ⇒ fixes κ_Z, assumption dropped

κ_c at 95% CL	observed	expected
On-shell, $\lvert\kappa_Z = \kappa_W\rvert \leq 1$	[−3.7, 3.2]	[−3.8, 3.3]
new On + off-shell, $\lvert\kappa_V\rvert \leq 1$	**[−3.3, 2.8]**	[−3.5, 2.9]
new On + off-shell, κ_Z, κ_W free	[−8.6, 7.5]	[−9.6, 8.3]

Higgs width: F. Errico (Tue) · p_T^H spectrum → backup



On-shell (blue, green), on + off-shell (red, black); solid observed, dashed expected

→ With $|\kappa_V| \leq 1$, H → 4ℓ matches the direct searches, under different assumptions

Where we stand

CMS constraints on κ_c at 95% CL

Probe	Reference	observed	expected
$t\bar{t}H$ + $VH(H \to c\bar{c})$	PRL 136 (2026) 011801	**$\lvert\kappa_c\rvert < 3.5$**	2.7
$t\bar{t}H(H \to c\bar{c})$	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.0$	3.3
$VH(H \to c\bar{c})$	PRL 131 (2023) 061801	$1.1 < \lvert\kappa_c\rvert < 5.5$	3.4
On + off-shell $H \to 4\ell$, $\lvert\kappa_V\rvert \leq 1$	arXiv:2607.23352	**[−3.3, 2.8]**	[−3.5, 2.9]
On-shell $H \to 4\ell$, $\lvert\kappa_Z = \kappa_W\rvert \leq 1$	PRD 112 (2025) 112001	[−3.7, 3.2]	[−3.8, 3.3]
On + off-shell $H \to 4\ell$, κ_Z, κ_W free	arXiv:2607.23352	[−8.6, 7.5]	[−9.6, 8.3]
cH ($\gamma\gamma$ + WW)	JHEP 08 (2026) 051	$\lvert\kappa_c\rvert < 47$	51
$H \to J/\psi\,\gamma$ · sign-sensitive · J. Bierkens, Thu	PLB 865 (2025) 139462	[−166, 208]	[−128, 169]

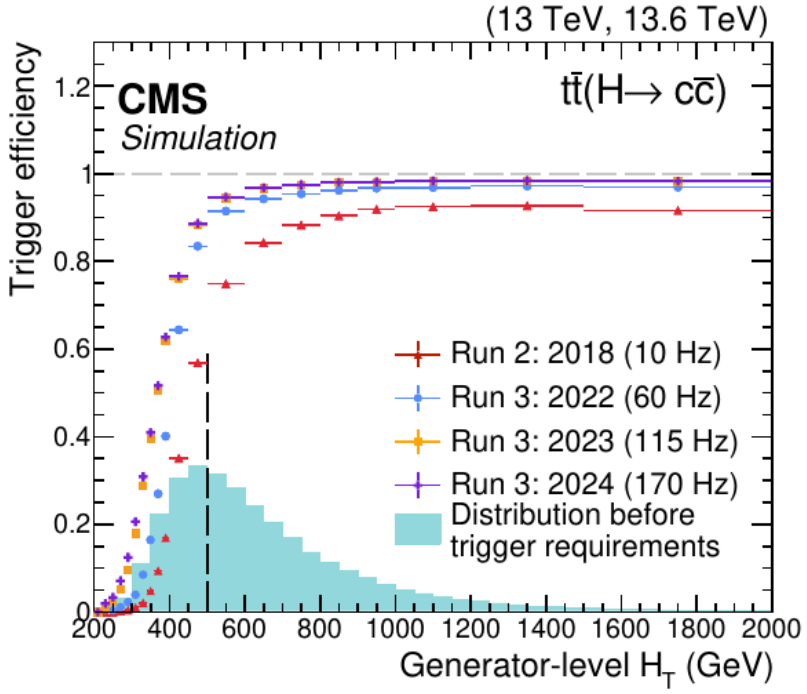
The results use different assumptions on the other Higgs couplings and are not directly comparable

→ Direct searches and on + off-shell $H \to 4\ell$ both reach $\lvert\kappa_c\rvert \sim 3$, where flavour-compatible BSM lives

Toward Run 3

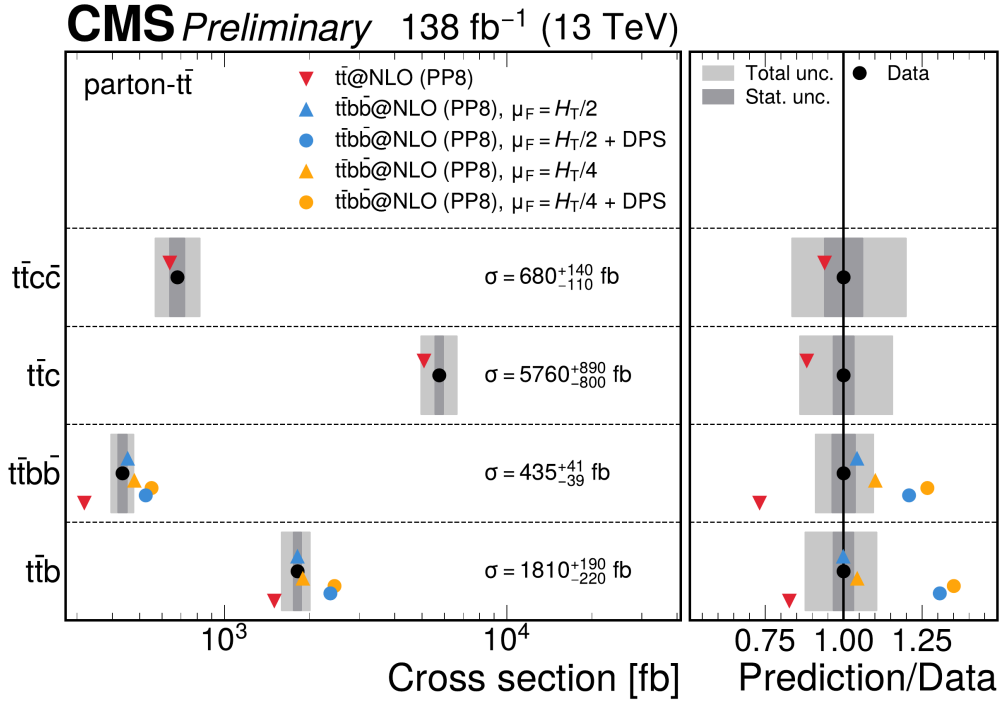
Better triggers, and the main $t\bar{t}H$ background measured

Run 3: ML b/c tagging at the HLT



arXiv:2608.18331 (BTV-25-002), simulation: $t\bar{t}H(H \to c\bar{c})$ trigger efficiency, Run 2 (2018, 13 TeV) vs Run 3 (2022–24, 13.6 TeV) · HLT taggers: J. Daniel (Mon) → backup

Run 2: $t\bar{t}c\bar{c}$ cross section measured



CMS-PAS-TOP-26-003 **new**, 138 fb^{-1}: $\sigma_{t\bar{t}c\bar{c}}$ = 680 ± 40 (stat) $^{+130}_{-100}$ (syst) fb

➜ Run 3 starts with higher trigger efficiency and a $t\bar{t}$+$c\bar{c}$ background measured in Run 2

Run 3 and HL-LHC prospects

arXiv:2504.00672 · CMS-HIG-25-002

From $|\kappa_c| \sim 3$ toward κ_c of order one

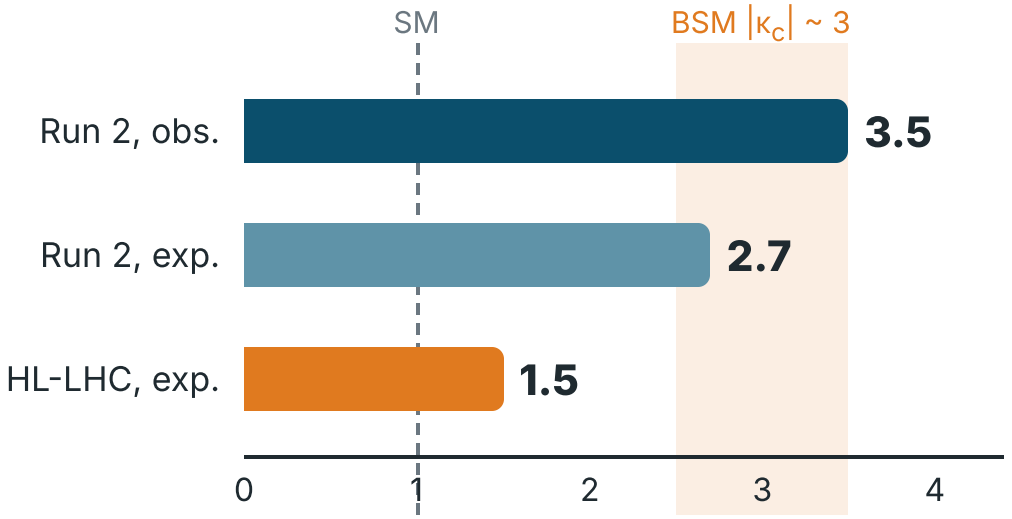
Run 3 (13.6 TeV)

- **Statistics-limited**: more data helps directly
- Better c tagging: UParT (slide 3)
- Better triggers (slide 17)

HL-LHC: 3000 fb^{-1} per experiment (≈ 20× Run 2)

- **Expected $|\kappa_c| < 1.5$**, ATLAS + CMS combined
- VH(H → $c\bar{c}$): ≈ 1.6σ expected, still no observation
- 30% better than the previous projection, thanks to better b and c tagging

Direct searches: allowed $|\kappa_c|$



Run 2: PRL 136 (2026) 011801 · HL-LHC: arXiv:2504.00672
No bar excludes $\kappa_c = 0$ yet

κ_c of order one: first sensitivity to whether the Higgs gives charm its mass

Summary

Does the Higgs give charm its mass?

The charm Yukawa is the most accessible second-generation quark coupling; CMS probes it **in the decay, in production and indirectly**

Direct

$t\bar{t}H$ + VH($H \to c\bar{c}$)

$|\kappa_c| < 3.5$ (2.7)

most stringent direct constraint, best expected sensitivity

Production

H + charm, γγ + WW

$|\kappa_c| < 47$ (51)

independent of the $H \to c\bar{c}$ decay

Indirect

on + off-shell $H \to 4\ell$

[−3.3, 2.8]

with $|\kappa_V| \leq 1$ (about ±8 without)

Next

Run 3 data

better taggers and triggers

$t\bar{t}c\bar{c}$ background already measured

Today **$|\kappa_c| \lesssim 3$–3.5**: entering BSM-motivated territory

HL-LHC expected **$|\kappa_c| < 1.5$**: approaching, not yet reaching, sensitivity to $\kappa_c = 0$

➜ Thank you!